

Parallels and contrasts between intermittently freezing and drying streams: From individual adaptations to biodiversity variation

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Abstract

1. Intermittently freezing and drying lotic systems are common in many parts of the world. These ecosystems provide dynamic habitats for biota, as both freezing and drying processes result in the loss of water flow along stream and river channels. However, research into the ecological effects of intermittent freezing on stream biota has remained relatively scarce compared to the recent increase in studies exploring the ecology of intermittent streams that dry.
2. Climate change is predicted to alter the distribution and dynamics of intermittently freezing stream types. A better understanding of the responses of stream biota to intermittent freezing would inform predictions of the effects of ongoing climate change on these ecosystems.
3. In this review, we compare the effects of both freezing and drying events on stream biota, ranging from individual-level adaptations to population effects, community-level biodiversity variation and food webs.
4. Despite major contrasts in environmental conditions (e.g. the drying stream channels are exposed to a direct interaction with the riparian zone during the dry phases and the freezing stream channels are enclosed during the freezing phases), we show that both freezing and drying phenomena have pervasive effects on the structure and dynamics of aquatic stream biota.
5. At the individual organism level, aquatic taxa use a range of physiological, morphological, life-history, and behavioural responses to persist despite changing habitat conditions in both freezing and drying streams. The variety of organisms' responses can result in both biodiversity increases and decreases, which may affect population and community dynamics in both ecosystem types.
6. Future research should further explore the population and community-level effects of freezing in stream ecosystems. Due to the ecological parallels between intermittently drying and freezing streams, approaches developed in research into drying streams may also be applied in freezing stream ecosystems. This could improve our understanding of climate change effects on freezing stream ecosystems.

KEYWORDS

ecological similarity, environmental harshness, ice, intermittency, lotic systems

1 | INTRODUCTION

Intermittent rivers and ephemeral streams (IRES; see Table 1 for further explanation of terms used in the review) that result from periodic drying of stream channels covers over 50% of the length of the global river network and are common in arid, semiarid, and Mediterranean climate regions (Datry, Larned, & Tockner, 2014). Past research on IRES has mainly focused on drying streams (Figure 1a), although, parallel to IRES, there are rivers and streams that sometimes freeze completely (Figure 1c). This phenomenon, in which all liquid water in a stream channel freezes to solid ice that reaches the streambed can be conceptualised as equivalent to the loss of surface water in drying IRES.

These freezing IRES (FIRES) are governed by seasonal freeze–thaw cycles (Allard, Buffin-Bélanger, & Bergeron, 2011). For example, it is estimated that 60% of river basins in the Northern Hemisphere experience some kind of freeze–thaw cycles during the winter (Allard et al., 2011), and the severity of these processes varies depending on changes in hydrological conditions, snowfall, and temperature (Shen, 2016). Whereas glacial streams are formed by channelised accumulation of water on, in, or under a glacier (Bennett & Glasser, 2010), FIRES can occur in unglaciated landscapes. The changing ice conditions in FIRES have direct and indirect effects on physical, chemical, and biological processes in freezing stream ecosystems (Prowse et al., 2006). However, based on extensive literature searches and personal observations by the authors, studies exploring the ecological effects of ice formation on freshwater ecosystems and their biodiversity are scarce when considering the generality of freezing processes at higher latitudes and altitudes in both hemispheres. In particular, only few papers (Olsson, 1981; Parker & Huryn, 2013) have examined streams that freeze totally down to the streambed. This lack of research on FIRES probably results from logistic difficulties in conducting research in streams during winter (e.g. due to low temperatures) and because of their remote geographical locations (e.g. in alpine environments and at high latitudes). However, a better understanding of the effects of freezing on stream biodiversity is urgently needed, because ongoing climate change is increasing temperatures, especially at high latitudes (Cohen et al., 2014), with the most pronounced changes expected to occur during winters (IPCC, 2013). These changes will alter ice conditions, such as the thickness and duration of ice cover, in stream ecosystems (IPCC, 2013; Stickler, Alfredsen, Linnansaari, & Fjeldstad, 2010). Evidence of postponed freezing and earlier thawing of ice cover has already been observed in high-latitude streams (Lind & Nilsson, 2015).

Because the loss of free water in both IRES and FIRES alters the connectivity and availability of suitable habitats for aquatic species, FIRES may experience similar ecological consequences to IRES that dry out completely (Figure 2). In recent years, an increasing body of literature has improved our knowledge of the ecology of streams that dry, helping researchers to predict biotic responses of IRES communities to drying (Datry, Bonada, & Boulton, 2017; Leigh et al.,

2016). However, FIRES remain a research challenge. Therefore, understanding the potential ecological similarities between intermittently freezing and drying streams could encourage researchers to apply approaches developed in IRES research to FIRES, and hence increase our knowledge of these freezing ecosystems.

In this review, we explore the parallels and contrasts between aquatic communities in FIRES and IRES ecosystems. We compare the effects of total freezing and total drying of stream channels on stream biota, ranging from individual organism-level adaptations to population dynamics, community-level biodiversity variation and food webs. We also highlight important FIRES research priorities.

2 | AQUATIC SPECIES HAVE DIVERSE ADAPTATIONS TO FREEZING AND DRYING

A range of physiological, morphological, life-history, and behavioural traits have been documented among aquatic species living in either freezing or drying streams. These traits can be divided into resistance (i.e. tolerating freezing or drying in situ) and resilience (i.e. avoiding freezing or drying by moving into a refuge or by recolonising the disturbed sites from refuges) responses, which allow individuals, populations, and communities to recover after a freezing or drying phase (Fritz & Dodds, 2004; Hershkovitz & Gasith, 2013; Strachan, Chester, & Robson, 2015; Townsend & Hildrew, 1994).

2.1 | Organism-level adaptations in cyanobacteria, algae, and plants

Streams that experience complete freezing or drying are occupied by species that have the appropriate traits and adaptations to prevail or avoid the extreme environmental conditions (Cavalli, Baattrup-Pedersen, & Riis, 2014; Southwood, 1977). Freshwater algae including chlorophytes and bacteria such as cyanophytes can survive periods of freezing due to resistant vegetative cells with thick cell walls, accumulation of reserve products within their cytoplasm, and lowered molecular weight solutes that lower their freezing point and prevent intracellular ice crystallisation (Hawes, 1990; Sheath, Vis, Hambrook, & Cole, 1996; Vincent & Howard-Williams, 1986). Some algal species also remain dormant in streambeds while waiting for the ice to thaw (Howard-Williams, Vincent, Broady, & Vincent, 1986). For example, Antarctic stream communities dominated by taxa adapted to severe freezing include cyanophytes such as *Phormidium*, *Oscillatoria*, *Nostoc*, and *Gloeocapsa*; chlorophytes such as *Binuclearia*, *Prasiola*, and *Tribonema*; and the pennate diatoms *Navicula* and *Hantzschia* (Howard-Williams et al., 1986).

Similar adaptations can be found in algae in drying streams, with algae such as the chlorophyte *Oedogonium* growing thickened cell walls and cystic cells to resist desiccation (Timoner, Buchaca, Acuña, & Sabater, 2014). Algal species can also have different growth forms in dry conditions, for example *Cymbella* and *Gomphonema* diatoms can form tubes or stalks that place cells within desiccation-tolerant filaments (Sabater et al., 2017). Some algae can also remain in the

TABLE 1 Glossary of the main terms used in the review

Term	Definition	References
Abbreviations		
IRES	Intermittent rivers and ephemeral streams: lotic systems that periodically cease to flow and may experience partial or complete streambed drying	Datry et al. (2017)
FIRES	Freezing intermittent rivers and ephemeral streams: lotic systems that periodically experience freezing of whole water column	
Other terms		
Anhydrobiosis	A state of reduced metabolism in which an organism loses most of its body water and produces substances that protect cells from desiccation-related damage. Normal metabolism resumes when favourable environmental conditions are restored	Guidetti and Jönsson (2002)
Antifreeze proteins	Substances that prevent the freezing of body fluids or tissues in sub-zero temperatures	Duman (2015)
Cryoprotectant	A substance that protects biological tissues and cells from the detrimental effects of freezing, mainly by preventing ice crystal formation	Karow (1969)
Diapause	A dormant state that arthropods enter in response to a predictable environmental stress. Metabolism slows and development ceases in response to external environmental signals and an internal biological clock. Development resumes when diapause is broken by an external stimulator such as increasing day length or specific temperatures	Gill, Goyal, and Chahil (2017)
Freeze-avoidant species	Species that avoid freezing behaviourally or by preventing their body fluids from freezing by using cryoprotectants and antifreeze proteins and by removing ice-nucleating agents through gut clearing	Frisbie and Lee (1997)
Freeze-tolerance species	Species that survive freezing of their body fluids by controlling extracellular ice formation by using ice-nucleating agents in their haemolymph, which promote freezing at relatively high sub-zero temperatures	Frisbie and Lee (1997)
Hyporheic zone	The sediments beneath and alongside a streambed, where shallow groundwater and surface water mix	Orghidan (2010)
Ice-nucleating agents	Minute solid particles or large molecules and proteins that promote formation of ice crystals by binding to water molecules	Duman (1982)
Inoculative freezing	Internal ice formation	Frisbie and Lee (1997)
Littoral	Shallow water zone near the edge of a river channel	Olsson (1981)
Quiescence	A response of an organism to an unpredictable environmental stress in which metabolism is reduced and development ceases	Gill et al. (2017)
Refuge	A place where a species can avoid detrimental environmental conditions	Bogan et al. (2015) and Huryn et al. (2005)
Reserve products	Components of algal cells such as starch and lipids, which support cell growth and survival by buffering variation in energy and carbon supplies	Vitova, Bisova, Kawano, and Zachleder (2015)
Supercooling	A process in which the temperature of liquids or gases lower below their freezing point without turning solid	Ashton (1983)

moist streambed sediments while waiting for water flow to resume (McKew, Taylor, McGenity, & Underwood, 2011).

To remain in their preferred habitat, some aquatic plants, such as pondweed (*Potamogeton*) species, can avoid being displaced by scouring ice through the formation of overwintering organs such as frost-tolerant or cold-resistant tubers, turions, and buds, or dormant apices and stems (Lind, Nilsson, Polvi, & Weber, 2014; Renman, 1989). In addition, vegetative growth and reproduction help plants to survive in frozen reaches (Lind et al., 2014; Nichols, Schloesser, & Hudson, 1989). Some vascular plants can even survive being enclosed in ice, as long as the basal parts of their rosettes remain unfrozen in the substrate (Renman, 1989).

In dry channels, plant persistence can also be facilitated by organs below the streambed surface. These plants then continue to grow from below-ground buds after flow resumes, even if their above-ground biomass decays during the dry period. Some aquatic plants further rely on diaspores and seed banks that begin to develop after water returns (Sabater et al., 2017). Physiological changes also promote resistance to drying in vascular plants. For instance, forget-me-nots (*Myosotis scorpioides*) can increase their leaf thickness (Germ & Gaberščik, 2003), and water lobelias (*Lobelia dortmanna*) can decrease the water content in their tissues to better withstand gravity changes in drying streams (Pedersen & Sand-Jensen, 1992; Sabater et al., 2017). Some plants, such as yellow water buttercup



FIGURE 1 Intermittently freezing and drying streams in different phases: the Audeux River, an intermittent stream during a dry phase (a) and a flowing phase (b); and a frozen forest stream under snow and ice cover (c). Photos a and b are from the Doubs, France; Credit: B. Launay. The frozen forest stream (c) is located in Oulu, Finland (Credit: K.E. Tolonen) [Colour figure can be viewed at wileyonlinelibrary.com]

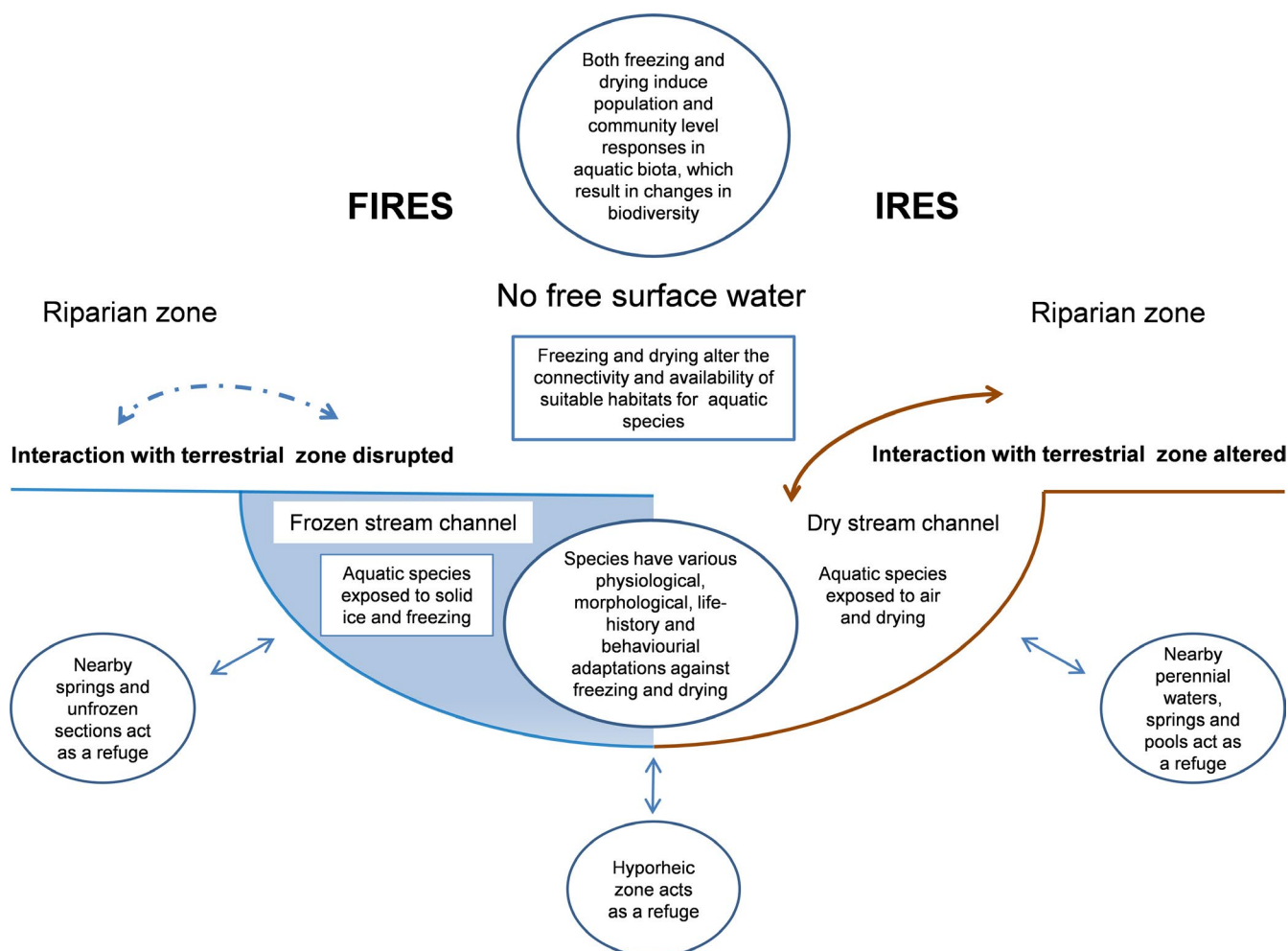


FIGURE 2 A schematic figure comparing how habitat connectivity and biota are altered by intermittency in freezing intermittent and ephemeral stream (FIRES) and in drying intermittent and ephemeral stream (IRES) channels [Colour figure can be viewed at wileyonlinelibrary.com]

(*Ranunculus flabellaris*) and the pondweed *Potamogeton wrightii*, can increase the thickness of their leaf cuticle or their stomatal density to regulate water and gas exchange (Sabater et al., 2017).

2.2 | Organism-level adaptations in animals

Like plants, animals living in either FIRES or IRES rely on various strategies to survive in their continuously changing habitats (Poff, 1997; Southwood, 1977). Physiological adaptations, such as freeze-avoidance and freeze-tolerance promote the resistance of organisms to ice, and often explain the ability of aquatic invertebrates to tolerate freezing conditions (Block, 1982; Danks, 2007; Irons, Miller, & Oswood, 1993; Oliver, 1968; Walters, Sformo, Barnes, & Duman, 2009). Macroinvertebrate survival is strongly influenced by direct contact with external ice, which induces faster and thus more adverse inoculative freezing, or avoidance of direct ice contact when supercooling enables them to withstand lower temperatures (Frisbie & Lee, 1997). Freeze-avoiding macroinvertebrates protect themselves by supercooling and maintaining their body fluids in a liquid state even below their natural freezing point. Species can lower their supercooling point by synthesising cryoprotectant and antifreeze proteins in the haemolymph, and by removing ice-nucleating agents through gut clearing (Lencioni, 2004). Freeze-tolerant species (e.g. the chironomids *Pseudodiamesa branickii* and *Diamesa cinerella*; Lencioni, Jousson, Guella, & Bernabò, 2015) can also tolerate extracellular ice formation through production of ice-nucleating agents in their haemolymph, which helps to control ice-crystal formation in their body fluids (Lee, 1989). This controlled extracellular ice formation helps to prevent cellular damage and provides time for metabolic adjustments to freezing (Block, Baust, Franks, Johnston, & Bale, 1990). For instance, the stonefly *Nemoura arctica* can survive being enclosed in ice by producing both glycerol and ice-binding factors (Walters et al., 2009). These molecular mechanisms, combined with an ability to migrate into the unfrozen hyporheic zone (Boulton, 2000; Orghidan, 2010), can even allow some macroinvertebrates to stay active in refuge areas during winter without entering a quiescent or diapausing state (Lencioni et al., 2015; Silveri, Tierno de Figueroa, & Maiolini, 2008).

It is also common for invertebrates to have life-history traits such as long life cycles, quiescence and diapause, which enable their development to extend over, and beyond, a cold season (Brittain, 1982; Danks, 2007). During quiescence or diapause, development is temporarily suspended, to tolerate adverse environmental conditions. Another frequently employed life-history adaptation that promotes resilience in cold environments is a reduction in the number of generations per year, and univoltinism or bivoltinism are favourable strategies in environments where the breeding season is very short (Danks, 1999; Lencioni, 2004).

For macroinvertebrate species that have adapted to withstand severe winter conditions enclosed in ice, the survival rates can be high (Mihalicz, 2015; Olsson, 1981; Walters et al., 2009). Olsson (1981) demonstrated a survival rate between 82 and 100% for individual macroinvertebrates enclosed in ice in the frozen substratum

of a littoral zone of a river. Further, living individuals of various worms (Nematoda, Oligochaeta), snails (Gastropoda), fingernail clams (Sphaeriidae), leeches (Hirudinea), waterlice (Isopoda), caddisflies (Trichoptera), and non-biting midges (Chironomidae) were found in the frozen substratum (Olsson, 1981). Many of the enclosed species, such as the snail *Gyraulus acronicus*, the worm *Lumbriculus variegatus* and some chironomid midges, had formed resting cysts, cocoons, and epiphragms that are not found when species are in an active state (Olsson, 1981). These constructions may enable the species to endure the mechanical stress exerted by ice (Danks, 2007). Other macroinvertebrate taxa, such as some mayfly (Ephemeroptera: Baetidae), stonefly (Plecoptera: Perlodidae), and blackfly (Diptera: Simuliidae) species, can also survive stream freezing in their egg stage (Clifford, 1969; Kurtak, 1974; Lillehammer, Brittain, Saltveit, & Nielsen, 1989).

Overall, these mechanisms resemble those that protect macroinvertebrates in drying IRES from desiccation (Bogan et al., 2017; Strachan et al., 2015). Invertebrate species can, for example, form different dormant stages to withstand dry conditions (Cover, Seo, & Resh, 2015; López-Rodríguez, Tierno de Figueroa, Fenoglio, Bo, & Alba-Tercedor, 2009). Species belonging to families such as the Enchytraeidae (Oligochaeta) can make desiccation-tolerant cocoons (Maraldo, Ravn, Slotsbo, & Holmstrup, 2009), whereas some copepods form resting-stage eggs when exposed to drying (Strachan et al., 2015). Some stoneflies and caddisflies can form desiccation-tolerant eggs in dry conditions (Strachan et al., 2015). Some IRES species can further survive long dry periods by entering an anhydrobiotic state in which dehydrated membranes and other cellular structures are stabilised by molecules, such as proteins and non-reducing sugars, which prevent desiccation from causing lethal structural damage (Crowe, Hoekstra, & Crowe, 1992; Strachan et al., 2015). Invertebrate species in IRES can also have different life-history adaptations in response to drying (López-Rodríguez, Tierno de Figueroa, & Alba-Tercedor, 2009). For instance, the stonefly *Brachyptera vera cordubensis* shows a fast univoltine life cycle with egg dormancy during dry phases. In contrast, the stonefly genus *Tyrrhenoleuctra* includes species that have a univoltine life cycle including a long adult flight period and nymphal dormancy during the dry phase. *Tyrrhenoleuctra* also shows signs of ovoviviparism in IRES, as its eggs are fully developed when laid, and hatch rapidly after laying when the flow resumes (López-Rodríguez, Tierno de Figueroa, & Alba-Tercedor, 2009).

The main behavioural adaptation promoting species resilience to freezing in running waters is to withdraw to habitats that do not freeze. For example, mayflies, stoneflies, and amphipods (Amphipoda) can be abundant in river margins during summer, but are not present there during winter, indicating that individuals have migrated from the freezing shoreline to unfrozen sections of the river (Olsson, 1981). Species may also avoid the freezing streambed by moving to nearby springs, where water generally remains ice-free and water temperatures are more stable due to the thermal influence of groundwater (Jones, Kielland, & Hinzman, 2015). Springs

also provide critical winter refuges, for example, for perleid stoneflies that withstand freezing (Huryn et al., 2005). Insects may also migrate vertically into the hyporheic zone of the streambed (Irons et al., 1993; Orghidan, 2010).

These behavioural responses are comparable to those of fauna in drying streams (Strachan et al., 2015), where species escape drying through means of dispersal such as drift, crawling and swimming (Boersma & Lytle, 2014; Paltridge, Dostine, Humphrey, & Boulton, 1997; Pires, Beja, & Magalhães, 2014; Robson, Matthews, Lind, & Thomas, 2008). Species can also burrow into the saturated subsurface sediments of the hyporheic zone (Kawanishi, Inoue, Dohi, Fujii, & Miyake, 2013; Timoner, Borrego, Acuña, & Sabater, 2014), although the usage of the hyporheic zone as a refuge from drying has also been questioned (del Rosario & Resh, 2000), and the use of the hyporheic zone can depend on the characteristics of the subsurface sediments and on the taxon in question (Stubbington, 2012). Species can also seek refuge in nearby permanent pools or perennial upstream and downstream reaches (Bogan, Boersma, & Lytle, 2015; Paltridge et al., 1997; Pires et al., 2014), from which recolonists arrive after flow returns. In addition, some species spend the dry period as aerial terrestrial forms (Bogan & Boersma, 2012). For instance, some stonefly and caddisfly species have aerial adults that aestivate in humid caves (Dobat, 1973; Salavert, Zamora-Muñoz, Ruiz-Rodríguez, Fernández-Cortés, & Soler, 2008).

3 | THE EFFECTS OF FREEZING AND DRYING ON POPULATIONS AND COMMUNITIES

3.1 | Influences on populations

Freezing and drying in lotic ecosystems can be major factors affecting aquatic organisms at the population level (Borgström & Museth, 2005). Complete freezing of streams can cause changes in population structure by affecting the recruitment of new individuals and by changing species distributions and abundances. Solid ice can limit the access of migratory fish to their spawning grounds in rivers (Bergeron, Roy, Chaumont, Mailhot, & Guay, 1998), or exclude fish from their summer habitats (Parker & Huryn, 2013). Streambed freezing can also affect the recruitment of young fish, such as brown trout (*Salmo trutta*), into local populations when small nursery streams with deposited eggs freeze completely (Borgström & Museth, 2005). However, in some cases, small fish such as young salmonids can use the hyporheic zone as a refuge at least during ice formation (Heggenes, Bremset, & Brabrand, 2010).

Drying causes fragmentation of populations and complicates fish migrations by disrupting migration pathways (Marshall et al., 2016; Pires et al., 2014). The hyporheic zone can support population maintenance of aquatic organisms in drying streams, by acting as a refuge for benthic fish and invertebrates following surface drying in IRES (Kawanishi et al., 2013; Vander Vorste, Corti, Sagouis, & Datry, 2016; Vander Vorste, Malard, & Datry, 2016). Drying events can further

affect population densities and population structure in IRES. For example, Pařil et al. (2019) demonstrated that short-term drying events severely reduce amphipod population densities and result in the loss of juveniles and reproductive females in the population in the first months after the flow resumed. The absence of juveniles was explained by their relatively weak dispersal capacities compared to faster and stronger swimming adults, which recolonised sites more quickly. The absence of reproductive females may reflect the physiological stress caused by drying, which could have caused early onset of reproductive diapause, as observed in response to other environmental stressors (Ladewig et al., 2006). Overall, such effects of drying could reduce long-term population persistence (Pařil et al., 2019).

3.2 | Influences on communities and biodiversity

Freezing and drying have both positive and negative effects on the biodiversity of stream communities. For instance, ice can serve directly as a habitat for algae and microbes during winter, hence increasing their total diversity (Frenette, Thibeault, Lapierre, & Hamilton, 2008). Further, ice-scouring can affect the abundance and distribution of aquatic plant and algae species by detaching them from the stream sediments (Rounick & Gregory, 1981; Scrimgeour, Prowse, Culp, & Chambers, 1994; Virtanen, Muotka, & Saksa, 2001), creating free habitat patches available for colonisation by additional species and thus increasing total species diversity (Lind et al., 2014; Virtanen et al., 2001). As a result, Virtanen et al. (2001) suggested that ice formation and scouring could be an important determinant of bryophyte community composition in northern streams, with frequent ice scouring preventing any species from becoming dominant and hence increasing local species diversity.

Substratum freezing can also affect the structure of communities (Parker & Huryn, 2011). When the streams freeze completely, communities can shift from a frost-sensitive to a frost-tolerant taxonomic composition (Erixon, 1979). Furthermore, Parker and Huryn (2011) compared the macroinvertebrate community structure of different stream types with different freezing propensities and identified distinct assemblages in different streams. In streams with evidence of substratum freezing, macroinvertebrate biomass mainly comprises species known to survive in frozen environments (Irons et al., 1993). In contrast, streams with perennial flow had the lowest proportion of biomass of freezing-tolerant taxa, whereas intermediate proportions of both types were found in streams where springs were present (Huryn et al., 2005). As a result, Huryn et al. (2005) suggested that the groundwater or hyporheic sources of the spring-fed streams may have provided a refuge from the freezing substratum (Orghidan, 2010).

In IRES, species richness and composition are also affected by temporal variability in the hydrological conditions (Dieterich & Anderson, 2000; Katz, Denslow, & Stromberg, 2012; Rolls, Heino, & Chessman, 2016). In general, drying has negative effects on local biodiversity (Datry, Larned, Fritz, et al., 2014; Datry, Larned, & Tockner, 2014; Timoner, Borrego, et al., 2014; Timoner, Buchaca, et al., 2014), and species in dry-phase aquatic communities are

typically dominated by subsets of communities present in streams during wet phases (Datry, Larned, Fritz, et al., 2014; Zeglin et al., 2011). However, drying IRES can have higher overall biodiversity due to change over time. Drying can allow channels to be temporarily colonised by terrestrial species, as well as by lentic taxa if pools remain within a largely dry channel (Bogan & Lytle, 2007; Corti & Datry, 2016; Stubbington, England, Wood, & Sefton, 2017). IRES can therefore support higher total species diversity (i.e. β and γ diversity) than aquatic habitats with more stable water conditions, due to lotic, lentic and, terrestrial taxa *taking turns* to occupy a single location (Bogan & Lytle, 2007; Katz et al., 2012; Sabater et al., 2017). Drying also induces changes in the structure of stream communities. For instance, Santos and Stevenson (2011) showed how the different flow regimes (perennial, intermittent, and ephemeral) in headwater streams supported distinct invertebrate community types, with a transition in functional feeding groups from perennial to ephemeral sites (Santos & Stevenson, 2011).

3.3 | Influences on food webs

Freezing of streams also affects the structure of food webs. For instance, ice cover blocks the light that streambed algal and microbial communities need to grow, hence affecting the primary production and autotrophy (Frenette et al., 2008). Parker and Huryn (2013) further examined how disturbance imposed by freezing structures food webs in FIRES of Arctic Alaska. Their results indicated that disturbances caused by bed movements and freezing resulted in low-complexity food webs (i.e. lower linkage densities and connectance) compared to communities in more stable streams (Parker & Huryn, 2013). The absence of fish also resulted in shorter food chain lengths in these streams. It was suggested that fish were excluded from some streams because of winter freezing and because of a lack of hydrological connections after spring thaw, which prevented colonisation during summer. Freezing can hence act as an environmental filter that affects community and food web structures in streams (Parker & Huryn, 2013; Scrimgeour et al., 1994).

Intermittency by drying also influences food webs in streams. For instance, changes in microbial communities during dry phases change the way in which detritus is processed (Datry et al., 2018), and food webs usually sustained by autotrophy during flowing phases moves to heterotrophy with drying (Reid, Quinn, Lake, & Reich, 2008). Drying also reduces the linkages, chain lengths, and connectivity of food webs in streams (Ledger, Brown, Edwards, Milner, & Woodward, 2013). Bogan and Lytle (2007) reported increases in the proportion of predators from 25% during high flows to 75% during low flows, increasing the predator–prey ratios as well as predator densities in remnant pools (Walters & Post, 2011). Overall, IRES food webs are characterised by smaller-sized consumers as large-bodied top predators tend to disappear during complete drying (Bogan & Lytle, 2011). Thus, food chain length in IRES becomes shorter compared to neighbouring perennial rivers (McHugh, Thompson, Greig, Warburton, & McIntosh, 2015). However, in contrast to FIRES where the ice cover interrupts the connection between streams and land

during winters, drying enables food webs to extend into terrestrial habitats (Corti & Datry, 2016; McIntosh et al., 2017). This gives riparian and terrestrial predators belonging to groups such as mammals, birds, lizards, and spiders the opportunity to eat aquatic organisms in drying pools (McIntosh et al., 2017). This increases the importance of terrestrial components in IRES food webs during dry phases (Boulton & Suter, 1986).

4 | SYNTHESIS AND AVENUES FOR FUTURE RESEARCH

In this review, we compare the effects of two types of surface water loss—drying and freezing—on individuals, populations, and communities of aquatic organisms in IRES and FIRES ecosystems (Table 2). Despite the fundamental environmental differences that distinguish drying streams, in which terrestrial habitats can extend into the channel, from freezing streams, in which ice creates a barrier between habitats, the loss of water affects the structure and dynamics of stream biota in both FIRES and IRES. Similar types of organism-level adaptations are used to ensure the long-term persistence of communities in both stream types. Responses are also observed at the population, community, and food-web levels. Although the mechanisms between the two stream types may differ, these different responses cause biodiversity to increase or decrease, affecting the dynamics of biota in both ecosystem types.

To date, ecological research on IRES has mainly focused on drying streams (Datry, Fritz, & Leigh, 2016; Leigh et al., 2016) and *freezing ecology* of aquatic environments in streams has not been thoroughly studied. In particular, studies on the ecological effects of total water column and bottom freezing in stream channels are scarce, making it difficult to make general conclusions about the effects of freezing on FIRES biodiversity. This lack of research in FIRES is concerning, as the percentage of river basins experiencing annual ice-processes in the Northern Hemisphere during winter is as high as 60% (Allard et al., 2011) and, just as drying affects ecosystem structure and functioning in IRES (von Schiller, Bernal, Dahm, & Martí, 2017), ice can have notable effects on physical and biological processes in FIRES (Allard et al., 2011; Prowse et al., 2006). Hence, more studies on the ecological effects of freezing on stream ecosystems should be implemented. This is even more urgent now that climate change-induced alterations in freezing patterns together with other anthropogenic stressors, such as increasing water abstraction (Rixen et al., 2011), will probably affect the winter habitat conditions for stream biota. For instance, intensive water abstraction for artificial snowmaking in ski resorts during natural low-flow periods in late autumn (de Jong, Lawler, & Essery, 2009) can lead to more frequent freezing–drying episodes. These changes could cause cascading and unpredictable effects throughout freezing aquatic ecosystems.

Changes in freezing–thawing cycles in FIRES due to climate change could alter their biodiversity, if species adapted to cope with certain ice conditions are faced with novel environments. Future climate scenarios predict that FIRES ecosystems will become less

TABLE 2 Summary table of the adaptations and responses of species, populations, communities, and food webs in freezing intermittent rivers and ephemeral streams (FIRES) and in drying intermittent rivers and ephemeral streams (IRES)

Level of organisation	Example organisms	Adaptations and responses		References
		FIRES	IRES	
Organisms	Algae, bacteria	Vegetative or dormant cells with thickened cell walls	Thickened cell walls and cystic cells	FIRES: Hawes (1990), Howard-Williams et al. (1986), Sheath et al. (1996), and Vincent and Howard-Williams (1986)
		Accumulation of reserve products within cytoplasm	Dormant cells	IRES: McKew et al. (2011) and Sabater et al. (2017)
		Lowered molecular weight solutes	Different growth forms	
	Aquatic plants	Overwintering organs such as turions, buds, or dormant apices and stems	Below-ground buds	FIRES: Lind et al. (2014), Nichols et al. (1989), and Renman (1989)
		Vegetative growth and reproduction	Diaspores and seed banks	
			Changes in tissue water content	IRES: Germ and Gaberšček (2003), Pedersen and Sand-Jensen (1992), and Sabater et al. (2017)
			Changes in leaf thickness	
	Invertebrates	Inoculative freezing	Anhydrobiotic state	FIRES: Block (1982), Boulton (2000), Danks (2007), Irons et al. (1993), Lencioni (2004), Oliver (1968), Olsson (1981), and Walters et al. (2009)
		Avoidance of direct ice contact	Univoltinism	
		Supercooling	Long adult flight period	
		Controlled extracellular ice formation	Nymphal dormancy	
		Long life-cycles	Ovoviviparism	
		Quiescence and diapause	Desiccation-tolerant cocoons, resting-stage eggs and desiccation-tolerant eggs	IRES: Bogan et al. (2017), Cover et al. (2015), Crowe et al. (1992), López-Rodríguez, Tierno de Figueroa, Fenoglio, et al. (2009), López-Rodríguez, Tierno de Figueroa, and Alba-Tercedor (2009), Maraldo et al. (2009), and Strachan et al. (2015)
Populations	Fish	Resting stages of cysts, cocoons and epiphragms		
		Movement to refuges such as springs and the hyporheic zone	Escape drying through means of dispersal such as drift, crawling and swimming to remnant pools, springs, hyporheic zone and other perennial reaches	FIRES: Hegggenes et al. (2010), Huryn et al. (2005), Irons et al. (1993), Olsson (1981), and Orghidan (2010)
		Changes in population structure by affecting the recruitment of new individuals	Aestivation in humid caves	IRES: Boersma and Lytle (2014), Bogan et al. (2015), Dobat (1973), Paltridge et al. (1997), Pires et al. (2014), Robson et al. (2008), and Salavert et al. (2008)
Populations	Fish	Difficulties for fish migrations	Difficulties for fish migrations	FIRES: Bergeron et al. (1998), Hegggenes et al. (2010), and Parker and Huryn (2013)
		Difficulties for fish migrations	Fragmentation of populations	IRES: Kawanishi et al. (2013), Marshall et al. (2016), Pařil et al. (2019), Pires et al. (2014), Vander Vorste, Corti, et al. (2016), and Vander Vorste, Malard, and Datry (2016)
		Fragmentation of populations	Altered population densities and population structures	

(Continues)

TABLE 2 (Continued)

Level of organisation	Adaptations and responses		References
	FIRES	IRES	
Communities	Example organisms Algae, aquatic plants, macroinvertebrates	Both positive and negative effects on biodiversity	FIRES: Frenette et al. (2008), Huryn et al. (2005), Parker and Huryn (2011), Rounick and Gregory (1981), Scrimgeour et al. (1994), and Virtanen et al. (2001)
		Changes in community composition and structure	IRES: Bogan and Lytle (2007), Detry, Larned, Fritz, et al. (2014), Dieterich and Anderson (2000), Katz et al. (2012), and Rolls et al. (2016)
Food webs		Changes in primary production	Timoner, Borrego, et al. (2014), Timoner, Buchaca, et al. (2014), and Sabater et al. (2017)
		Lower levels of linkage densities and connections	FIRES: Frenette et al. (2008), Parker and Huryn (2013), and Scrimgeour et al. (1994) IRES: Ledger et al. (2013) and Reid et al. (2008)

fragmented and warmer, and, if biota cannot adapt to such changes, the structure of ecological communities in freezing streams could experience substantial alterations. Comparative studies of the potential effects of changing habitat dynamics on multiple biological groups in FIRES are needed. Because of the similarities between intermittently freezing and drying streams, applying approaches developed in IRES research, and comparing findings from IRES with those from FIRES could increase our knowledge of these overlooked freezing ecosystems. A better understanding of FIRES ecology will in turn help us to predict the effects of climate change on these ecosystems and manage them efficiently in the future.

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REFERENCES

- Allard, G., Buffin-Bélanger, T., & Bergeron, N. (2011). Analysis of frazil ice as a geomorphic agent in a frazil pool. *River Research and Applications*, 27(9), 1136–1148. <https://doi.org/10.1002/rra.1490>
- Ashton, G. D. (1983). Frazil ice. In R. E. Meyer (Ed.), *Theory of dispersed multiphase flow* (1st ed., pp. 271–289). Cambridge, MA: Academic Press. <https://doi.org/10.1016/b978-0-12-493120-6.50017-9>
- Bennett, M., & Glasser, N. (2010). *Glacial geology: ice sheets and landforms* (2nd ed.). Chichester, UK: John Wiley & Sons.
- Bergeron, N. E., Roy, A. G., Chaumont, D., Mailhot, Y., & Guay, É. (1998). Winter geomorphological processes in the Sainte-Anne River (Québec) and their impact on the migratory behaviour of Atlantic tomcod (*Microgadus tomcod*). *Regulated Rivers: Research & Management*, 14(1), 95–105. [https://doi.org/10.1002/\(sici\)1099-1646\(199801/02\)14:1<95:aid-rrr481>3.0.co;2-2](https://doi.org/10.1002/(sici)1099-1646(199801/02)14:1<95:aid-rrr481>3.0.co;2-2)
- Block, W. (1982). Supercooling points of insects and mites on the Antarctic Peninsula. *Ecological Entomology*, 7(1), 1–8. <https://doi.org/10.1111/j.1365-2311.1982.tb00638.x>
- Block, W., Baust, J. G., Franks, F., Johnston, I. A., & Bale, J. (1990). Cold tolerance of insects and other arthropods [and discussion]. *Philosophical Transactions of the Royal Society B: Biological Sciences*, 326(1237), 613–633. <https://doi.org/10.1098/rstb.1990.0035>
- Boersma, K. S., & Lytle, D. A. (2014). Overland dispersal and drought-escape behavior in a flightless aquatic insect, *Abedus herberti* (Hemiptera: Belostomatidae). *Southwestern Naturalist*, 59(2), 301–302. <https://doi.org/10.1894/n09-frg-07.1>
- Bogan, M. T., & Boersma, K. S. (2012). Aerial dispersal of aquatic invertebrates along and away from arid-land streams. *Freshwater Science*, 31(4), 1131–1144. <https://doi.org/10.1899/12-066.1>

- Bogan, M. T., Boersma, K. S., & Lytle, D. A. (2015). Resistance and resilience of invertebrate communities to seasonal and supraseasonal drought in arid-land headwater streams. *Freshwater Biology*, 60(12), 2547–2558. <https://doi.org/10.1111/fwb.12522>
- Bogan, M. T., Chester, E. T., Datry, T., Murphy, A. L., Robson, B. J., Ruhi, A., ... Whitney, J. E. (2017). Resistance, resilience, and community recovery in intermittent rivers and ephemeral streams. In T. Datry, N. Bonada, & A. Boulton (Eds.), *Intermittent rivers and ephemeral streams: Ecology and management* (pp. 349–376). Burlington, VT: Elsevier. <https://doi.org/10.1016/b978-0-12-803835-2.00013-9>
- Bogan, M. T., & Lytle, D. A. (2007). Seasonal flow variation allows “time-sharing” by disparate aquatic insect communities in montane desert streams. *Freshwater Biology*, 52(2), 290–304. <https://doi.org/10.1111/j.1365-2427.2006.01691.x>
- Bogan, M. T., & Lytle, D. A. (2011). Severe drought drives novel community trajectories in desert stream pools. *Freshwater Biology*, 56(10), 2070–2081. <https://doi.org/10.1111/j.1365-2427.2011.02638.x>
- Borgström, R., & Museth, J. (2005). Accumulated snow and summer temperature – Critical factors for recruitment to high mountain populations of brown trout (*Salmo trutta* L.). *Ecology of Freshwater Fish*, 14(4), 375–384. <https://doi.org/10.1111/j.1600-0633.2005.00112.x>
- Boulton, A. J. (2000). The functional role of the hyporheos. *Internationale Vereinigung für Theoretische und Angewandte Limnologie: Verhandlungen*, 27, 51–63. <https://doi.org/10.1080/03680770.1998.11901203>
- Boulton, A. J., & Suter, P. J. (1986). Ecology of temporary streams – An Australian perspective. In P. De Deckker & W. Williams (Eds.), *Limnology in Australia* (pp. 313–327). Melbourne, Vic./Dordrecht, the Netherlands: Springer. https://doi.org/10.1007/978-94-009-4820-4_19
- Brittain, J. E. (1982). Biology of mayflies. *Annual Review of Entomology*, 27, 119–147. <https://doi.org/10.1146/annurev.en.27.010182.001003>
- Cavalli, G., Baattrup-Pedersen, A., & Riis, T. (2014). The role of species functional traits in distributional patterns of lowland stream vegetation. *Freshwater Science*, 33(4), 1074–1085. <https://doi.org/10.1086/678048>
- Clifford, H. F. (1969). Limnological features of a northern brown-water stream, with special reference to the life histories of the aquatic insects. *The American Midland Naturalist*, 82(2), 578–597. <https://doi.org/10.2307/2423797>
- Cohen, J., Screen, J. A., Furtado, J. C., Barlow, M., Whittleston, D., Coumou, D., ... Jones, J. (2014). Recent Arctic amplification and extreme mid-latitude weather. *Nature Geoscience*, 7(9), 627–637. <https://doi.org/10.1038/ngeo2234>
- Corti, R., & Datry, T. (2016). Terrestrial and aquatic invertebrates in the riverbed of an intermittent river: Parallels and contrasts in community organisation. *Freshwater Biology*, 61(8), 1308–1320. <https://doi.org/10.1111/fwb.12692>
- Cover, M. R., Seo, J. H., & Resh, V. H. (2015). Life history, burrowing behavior, and distribution of *Neohermes filicornis* (Megaloptera: Corydalidae), a long-lived aquatic insect in intermittent streams. *Western North American Naturalist*, 75(4), 474–490. <https://doi.org/10.3398/064.075.0405>
- Crowe, J. H., Hoekstra, F. A., & Crowe, L. M. (1992). Anhydrobiosis. *Annual Review of Physiology*, 54, 579–599. <https://doi.org/10.1146/annurev.ph.54.030192.003051>
- Danks, H. V. (1999). Life cycles in polar arthropods – Flexible or programmed? *European Journal of Entomology*, 96(2), 83–102. <https://doi.org/10.14411/eje.1999.018.html>
- Danks, H. V. (2007). How aquatic insects live in cold climates. *The Canadian Entomologist*, 139(4), 443–471. <https://doi.org/10.4039/n06-100>
- Datry, T., Bonada, N., & Boulton, A. (2017). General introduction. In T. Datry, N. Bonada, & A. Boulton (Eds.), *Intermittent rivers and ephemeral streams: Ecology and management* (1st ed., pp. 1–20). Burlington, VT: Elsevier.
- Datry, T., Foulquier, A., Corti, R., von Schiller, D., Tockner, K., Mendoza-Lera, C., ... Zoppini, A. (2018). A global analysis of terrestrial plant litter dynamics in non-perennial waterways. *Nature Geoscience*, 11, 497–503. <https://doi.org/10.1038/s41561-018-0134-4>
- Datry, T., Fritz, K., & Leigh, C. (2016). Challenges, developments and perspectives in intermittent river ecology. *Freshwater Biology*, 61(8), 1171–1180. <https://doi.org/10.1111/fwb.12789>
- Datry, T., Larned, S. T., Fritz, K. M., Bogan, M. T., Wood, P. J., Meyer, E. I., & Santos, A. N. (2014). Broad-scale patterns of invertebrate richness and community composition in temporary rivers: Effects of flow intermittence. *Ecography*, 37(1), 94–104. <https://doi.org/10.1111/j.1600-0587.2013.00287.x>
- Datry, T., Larned, S. T., & Tockner, K. (2014). Intermittent rivers: A challenge for freshwater ecology. *BioScience*, 64(3), 229–235. <https://doi.org/10.1093/biosci/bit027>
- de Jong, C., Lawler, D., & Essery, R. (2009). Mountain hydroclimatology and snow seasonality: Perspectives on climate impacts, snow seasonality and hydrological change in mountain environments. *Hydrological Processes*, 23(7), 955–961. <https://doi.org/10.1002/hyp.7193>
- del Rosario, R. B., & Resh, V. H. (2000). Invertebrates in intermittent and perennial streams: Is the hyporheic zone a refuge from drying? *Journal of the North American Benthological Society*, 19(4), 680–696. <https://doi.org/10.2307/1468126>
- Dieterich, M., & Anderson, N. H. (2000). The invertebrate fauna of summer-dry streams in western Oregon. *Fundamental and Applied Limnology*, 147(3), 273–295. <https://doi.org/10.1127/archiv-hydro-biol/147/2000/273>
- Dobat, K. (1973). Observations on marked and unmarked Trichoptera in the Barehole in Lonetal (Swabian Jura). *International Journal of Speleology*, 5(1), 63–85. Retrieved from <http://www.speleogenesis.info/directory/karstbase/publication.php?xml:id=5179>
- Duman, J. G. (1982). Insect antifreezes and ice-nucleating agents. *Cryobiology*, 19(6), 613–627. [https://doi.org/10.1016/0011-2240\(82\)90191-2](https://doi.org/10.1016/0011-2240(82)90191-2)
- Duman, J. G. (2015). Animal ice-binding (antifreeze) proteins and glycolipids: An overview with emphasis on physiological function. *The Journal of Experimental Biology*, 218(12), 1846–1855. <https://doi.org/10.1242/jeb.116905>
- Erixon, G. (1979). Environment and aquatic vegetation of a riverside lagoon in Northern Sweden. *Aquatic Botany*, 6, 95–109. [https://doi.org/10.1016/0304-3770\(79\)90054-8](https://doi.org/10.1016/0304-3770(79)90054-8)
- Frenette, J. J., Thibeault, P., Lapierre, J.-F., & Hamilton, P. B. (2008). Presence of algae in freshwater ice cover of fluvial lac Saint-Pierre (St. Lawrence River, Canada). *Journal of Phycology*, 44(2), 284–291. <https://doi.org/10.1111/j.1529-8817.2008.00481.x>
- Frisbie, M. P., & Lee, R. E. (1997). Inoculative freezing and the problem of winter survival for freshwater macroinvertebrates. *Journal of the North American Benthological Society*, 16(3), 635–650. <https://doi.org/10.2307/1468150>
- Fritz, K. M., & Dodds, W. K. (2004). Resistance and resilience of macroinvertebrate assemblages to drying and flood in a tallgrass prairie stream system. *Hydrobiologia*, 527(1), 99–112. <https://doi.org/10.1023/b:hydr.0000043188.53497.9b>
- Germ, M., & Gaberšček, A. (2003). Comparison of aerial and submerged leaves in two amphibious species, *Myosotis scorpioides* and *Ranunculus trichophyllus*. *Photosynthetica*, 41(1), 91–96. <https://doi.org/10.1023/a:1025816614572>
- Gill, H. K., Goyal, G., & Chahil, G. (2017). Insect diapause: A review. *Journal of Agricultural Science and Technology A*, 7, 454–473. <https://doi.org/10.17265/2161-6256/2017.07.002>
- Guidetti, R., & Jönsson, K. I. (2002). Long-term anhydrobiotic survival in semi-terrestrial micrometazoans. *Journal of Zoology*, 257(2), 181–187. <https://doi.org/10.1017/s095283690200078x>

- Hawes, I. (1990). Effects of freezing and thawing on a species of *Zygnema* (Chlorophyta) from the Antarctic. *Phycologia*, 29(3), 326–331. <https://doi.org/10.2216/i0031-8884-29-3-326.1>
- Heggnes, J., Bremset, G., & Brabrand, Å. (2010). Groundwater, critical habitats, and behaviour of Atlantic salmon, brown trout and Arctic char in streams – NINA Report 654. Trondheim.
- Hershkovitz, Y., & Gasith, A. (2013). Resistance, resilience, and community dynamics in Mediterranean-climate streams. *Hydrobiologia*, 719(1), 59–75. <https://doi.org/10.1007/s10750-012-1387-3>
- Howard-Williams, C., Vincent, C. L., Broady, P. A., & Vincent, W. F. (1986). Antarctic stream ecosystems: Variability in environmental properties and algal community structure. *Hydrobiology*, 71(4), 511–544. <https://doi.org/10.1002/iroh.19860710405>
- Huryn, A. D., Slavik, K. A., Lowe, R. L., Parker, S. M., Anderson, D. S., & Peterson, B. J. (2005). Landscape heterogeneity and the biodiversity of Arctic stream communities: A habitat template analysis. *Canadian Journal of Fisheries and Aquatic Sciences*, 62(8), 1905–1919. <https://doi.org/10.1139/f05-100>
- IPCC (2013). Climate Change 2013: The physical science basis. Contribution of working group I to the Fifth assessment report of the Intergovernmental panel on climate change. Cambridge, United Kingdom and New York, USA.
- Irons, J. G. III, Miller, L. K., & Oswood, M. W. (1993). Ecological adaptations of aquatic macroinvertebrates to overwintering in interior Alaska (U.S.A.) subarctic streams. *Canadian Journal of Zoology*, 71(1), 98–108. <https://doi.org/10.1139/z93-015>
- Jones, C., Kielland, K., & Hinzman, L. (2015). Modeling groundwater upwelling as a control on river ice thickness. *Hydrology Research*, 46(4), 566–577. <https://doi.org/10.2166/nh.2015.026>
- Karow, A. M. (1969). Cryoprotectants—a new class of drugs. *Journal of Pharmacy and Pharmacology*, 21(4), 209–223. <https://doi.org/10.1111/j.2042-7158.1969.tb08235.x>
- Katz, G. L., Denslow, M. W., & Stromberg, J. C. (2012). The Goldilocks effect: Intermittent streams sustain more plant species than those with perennial or ephemeral flow. *Freshwater Biology*, 57(3), 467–480. <https://doi.org/10.1111/j.1365-2427.2011.02714.x>
- Kawanishi, R., Inoue, M., Dohi, R., Fujii, A., & Miyake, Y. (2013). The role of the hyporheic zone for a benthic fish in an intermittent river: A refuge, not a graveyard. *Aquatic Sciences*, 75(3), 425–431. <https://doi.org/10.1007/s00027-013-0289-4>
- Kurtak, D. (1974). Overwintering of *Simulium pictipes* Hagen (Diptera: Simuliidae) as eggs. *Journal of Medical Entomology*, 11(3), 383–384. <https://doi.org/10.1093/jmedent/11.3.383>
- Ladewig, V., Jungmann, D., Köhler, H. R., Schirling, M., Triebkorn, R., & Nagel, R. (2006). Population structure and dynamics of *Gammarus fossarum* (Amphipoda) upstream and downstream from effluents of sewage treatment plants. *Archives of Environmental Contamination and Toxicology*, 50(3), 370–383. <https://doi.org/10.1007/s00244-005-7039-0>
- Ledger, M. E., Brown, L. E., Edwards, F. K., Milner, A. M., & Woodward, G. (2013). Drought alters the structure and functioning of complex food webs. *Nature Climate Change*, 3(3), 223–227. <https://doi.org/10.1038/nclimate1684>
- Lee, R. E. (1989). Insect cold-hardiness: To freeze or not to freeze. *BioScience*, 39(5), 308–313. <https://doi.org/10.2307/1311113>
- Leigh, C., Boulton, A. J., Courtwright, J. L., Fritz, K., May, C. L., Walker, R. H., & Datry, T. (2016). Ecological research and management of intermittent rivers: An historical review and future directions. *Freshwater Biology*, 61(8), 1181–1199. <https://doi.org/10.1111/fwb.12646>
- Lencioni, V. (2004). Survival strategies of freshwater insects in cold environments. *Journal of Limnology*, 63(1), 45–55. <https://doi.org/10.4081/jlimnol.2004.s1.45>
- Lencioni, V., Jousson, O., Guella, G., & Bernabò, P. (2015). Cold adaptive potential of chironomids overwintering in a glacial stream. *Physiological Entomology*, 40(1), 43–53. <https://doi.org/10.1111/phen.12084>
- Lillehammer, A., Brittain, J. E., Saltveit, S. J., & Nielsen, P. S. (1989). Egg development, nymphal growth and life cycle strategies in plecoptera. *Holarctic Ecology*, 12(2), 173–186. Retrieved from <http://www.jstor.org/stable/3682202>
- Lind, L., & Nilsson, C. (2015). Vegetation patterns in small boreal streams relate to ice and winter floods. *Journal of Ecology*, 103(2), 431–440. <https://doi.org/10.1111/1365-2745.12355>
- Lind, L., Nilsson, C., Polvi, L. E., & Weber, C. (2014). The role of ice dynamics in shaping vegetation in flowing waters. *Biological Reviews*, 89(4), 791–804. <https://doi.org/10.1111/brv.12077>
- López-Rodríguez, M. J., Tiernode Figueroa, J. M., & Alba-Tercedor, J. (2009). Life history, feeding and secondary production of two Nemouroidea species (Plecoptera, Insecta) in a temporary stream of the Southern Iberian Peninsula. *Fundamental and Applied Limnology*, 175(2), 161–170. <https://doi.org/10.1127/1863-9135/2009/0175-0161>
- López-Rodríguez, M. J., Tierno de Figueroa, J. M., Fenoglio, S., Bo, T., & Alba-Tercedor, J. (2009). Life strategies of 3 Perlodidae species (Plecoptera) in a Mediterranean seasonal stream in southern Europe. *Journal of the North American Benthological Society*, 28(3), 611–625. <https://doi.org/10.1899/08-105.1>
- Maraldo, K., Ravn, H. W., Slotsbo, S., & Holmstrup, M. (2009). Responses to acute and chronic desiccation stress in *Enchytraeus* (Oligochaeta: Enchytraeidae). *Journal of Comparative Physiology B*, 179(2), 113–123. <https://doi.org/10.1007/s00360-008-0305-5>
- Marshall, J. C., Menke, N., Crook, D. A., Lobeggeiger, J. S., Balcombe, S. R., Huey, J. A., ... Arthington, A. H. (2016). Go with the flow: The movement behaviour of fish from isolated waterhole refugia during connecting flow events in an intermittent dryland river. *Freshwater Biology*, 61(8), 1242–1258. <https://doi.org/10.1111/fwb.12707>
- McHugh, P. A., Thompson, R. M., Greig, H. S., Warburton, H. J., & McIntosh, A. R. (2015). Habitat size influences food web structure in drying streams. *Ecography*, 38(7), 700–712. <https://doi.org/10.1111/ecog.01193>
- McIntosh, A. R., Leigh, C., Boersma, K. S., McHugh, P. A., Febria, C., & García-Berthou, E. (2017). Food webs and trophic interactions in intermittent rivers and ephemeral streams. In T. Datry, N. Bonada, & A. Boulton (Eds.), *Intermittent rivers and ephemeral streams: Ecology and management* (1st ed., pp. 323–347). Burlington, VT: Elsevier.
- McKew, B. A., Taylor, J. D., McGenity, T. J., & Underwood, G. J. C. (2011). Resistance and resilience of benthic biofilm communities from a temperate saltmarsh to desiccation and rewetting. *The ISME Journal*, 5(1), 30–41. <https://doi.org/10.1038/ismej.2010.91>
- Mihalic, J. E. (2015). An observation of the overwintering aquatic insects in a Prairie Pond in Saskatchewan, Canada. *USURJ: University of Saskatchewan Undergraduate Research Journal*, 1(2), 99–106.
- Nichols, S. J., Schloesser, D. W., & Hudson, P. L. (1989). Submersed macrophyte communities before and after an episodic ice jam in the St. Clair and Detroit rivers. *Canadian Journal of Botany*, 67(8), 2364–2370. <https://doi.org/10.1139/b89-302>
- Oliver, D. R. (1968). Adaptations of arctic chironomidae. *Annales Zoologici Fennici*, 5(1), 111–118. Retrieved from <http://www.jstor.org/stable/23731452>
- Olsson, T. I. (1981). Overwintering of benthic macroinvertebrates in ice and frozen sediment in a North Swedish river. *Holarctic Ecology*, 4(3), 161–166. Retrieved from <http://www.jstor.org/stable/3682535>
- Orghidan, T. (2010). A new habitat of subsurface waters: The hyporheic biotope. *Fundamental and Applied Limnology*, 176(4), 291–302. <https://doi.org/10.1127/1863-9135/2010/0176-0291>
- Paltridge, R., Dostine, P., Humphrey, C., & Boulton, A. (1997). Macroinvertebrate recolonization after re-wetting of a tropical seasonally-flowing stream (Magela Creek, Northern Territory, Australia).

- Marine and Freshwater Research*, 48(7), 633–645. <https://doi.org/10.1071/mf97059>
- Pařil, P., Leigh, C., Polášek, M., Sarremejane, R., Řezníčková, P., Dostálová, A., & Stubbington, R. (2019). Short-term streambed drying events alter amphipod population structure in a central European stream. *Fundamental and Applied Limnology*, <https://doi.org/10.1127/fal/2019/1164>
- Parker, S. M., & Huryn, A. D. (2011). Effects of natural disturbance on stream communities: A habitat template analysis of arctic headwater streams. *Freshwater Biology*, 56(7), 1342–1357. <https://doi.org/10.1111/j.1365-2427.2011.02573.x>
- Parker, S. M., & Huryn, A. D. (2013). Disturbance and productivity as co-determinants of stream food web complexity in the Arctic. *Limnology and Oceanography*, 58(6), 2158–2170. <https://doi.org/10.4319/lo.2013.58.6.2158>
- Pedersen, O., & Sand-Jensen, K. (1992). Adaptations of submerged *Lobelia dortmanna* to aerial life form: Morphology, carbon sources and oxygen dynamics. *Oikos*, 65(1), 89. <https://doi.org/10.2307/3544890>
- Pires, D. F., Beja, P., & Magalhães, M. F. (2014). Out of pools: Movement patterns of Mediterranean stream fish in relation to dry season refugia. *River Research and Applications*, 30(10), 1269–1280. <https://doi.org/10.1002/rra.2776>
- Poff, N. L. (1997). Landscape filters and species traits: Towards mechanistic understanding and prediction in stream ecology. *Journal of the North American Benthological Society*, 16(2), 391–409. <https://doi.org/10.2307/1468026>
- Prowse, T. D., Wrona, F., Reist, J., Hobbie, J., Lévesque, L., & Vincent, W. (2006). General features of the arctic relevant to climate change in freshwater ecosystems. *Ambio*, 35(7), 330–338. [https://doi.org/10.1579/0044-7447\(2006\)35\[330:GFOTAR\]2.0.CO;2](https://doi.org/10.1579/0044-7447(2006)35[330:GFOTAR]2.0.CO;2)
- Reid, D. J., Quinn, G. P., Lake, P. S., & Reich, P. (2008). Terrestrial detritus supports the food webs in lowland intermittent streams of south-eastern Australia: A stable isotope study. *Freshwater Biology*, 53(10), 2036–2050. <https://doi.org/10.1111/j.1365-2427.2008.02025.x>
- Renman, G. (1989). Distribution of littoral macrophytes in a North Swedish riverside lagoon in relation to bottom freezing. *Aquatic Botany*, 33(3–4), 243–256. [https://doi.org/10.1016/0304-3770\(89\)90040-5](https://doi.org/10.1016/0304-3770(89)90040-5)
- Rixen, C., Teich, M., Lardelli, C., Gallati, D., Pohl, M., Pütz, M., & Bebi, P. (2011). Winter tourism and climate change in the Alps: An assessment of resource consumption, snow reliability, and future snowmaking potential. *Mountain Research and Development*, 31(3), 229–236. <https://doi.org/10.1659/mrd-journal-d-10-00112.1>
- Robson, B. J., Matthews, T. G., Lind, P. R., & Thomas, N. A. (2008). Pathways for algal recolonization in seasonally-flowing streams. *Freshwater Biology*, 53(12), 2385–2401. <https://doi.org/10.1111/j.1365-2427.2008.02061.x>
- Rolls, R. J., Heino, J., & Chessman, B. C. (2016). Unravelling the joint effects of flow regime, climatic variability and dispersal mode on beta diversity of riverine communities. *Freshwater Biology*, 61(8), 1350–1364. <https://doi.org/10.1111/fwb.12793>
- Rounick, J. S., & Gregory, S. V. (1981). Temporal changes in periphyton standing crop during an unusually dry winter in streams of the Western Cascades, Oregon. *Hydrobiologia*, 83(2), 197–205. <https://doi.org/10.1007/bf00008267>
- Sabater, S., Bornette, G., De Wilde, M., Timoner, X., Stromberg, J., & Stella, J. C. (2017). The biota of intermittent rivers: Algae and vascular plants. In T. Datry, N. Bonada, & A. Boulton (Eds.), *Intermittent rivers and ephemeral streams: Ecology and management* (1st ed., pp. 189–216). Burlington, VT: Elsevier.
- Salavert, V., Zamora-Muñoz, C., Ruiz-Rodríguez, M., Fernández-Cortés, A., & Soler, J. J. (2008). Climatic conditions, diapause and migration in a troglophile caddisfly. *Freshwater Biology*, 53(8), 1606–1617. <https://doi.org/10.1111/j.1365-2427.2008.02000.x>
- Santos, A. N., & Stevenson, R. D. (2011). Comparison of macroinvertebrate diversity and community structure among perennial and non-perennial headwater streams. *Northeastern Naturalist*, 18(1), 7–26. <https://doi.org/10.1656/045.018.0102>
- Scrimgeour, G. J., Prowse, T. D., Culp, J. M., & Chambers, P. A. (1994). Ecological effects of river ice break-up: A review and perspective. *Freshwater Biology*, 32(2), 261–275. <https://doi.org/10.1111/j.1365-2427.1994.tb01125.x>
- Sheath, R. G., Vis, M. L., Hambrook, J. A., & Cole, K. M. (1996). Tundra stream macroalgae of North America: Composition, distribution and physiological adaptations. *Biogeography of Freshwater Algae*, 118, 67–82. https://doi.org/10.1007/978-94-017-0908-8_6
- Shen, H. T. (2016). River ice processes. In L. Wang, C. Yang, & M. Wang (Eds.), *Advances in water resources management* (pp. 483–530). Cham, Switzerland: Springer.
- Silveri, L., Tierno de Figueroa, J. M., & Maiolini, B. (2008). Feeding habits of Perlodidae (Plecoptera) in the hyporheic habitats of Alpine streams (Trentino-NE Italy). *Entomologica Fennica*, 19, 176–183.
- Southwood, T. (1977). Habitat, the templet for ecological strategies? *Journal of Animal Ecology*, 46(2), 337–365. <https://doi.org/10.2307/3817>
- Stickler, M., Alfredsen, K. T., Linnansaari, T., & Fjeldstad, H. (2010). The influence of dynamic ice formation on hydraulic heterogeneity in steep streams. *River Research and Applications*, 26(9), 1187–1197. <https://doi.org/10.1002/rra.1331>
- Strachan, S. R., Chester, E. T., & Robson, B. J. (2015). Freshwater invertebrate life history strategies for surviving desiccation. *Springer Science Reviews*, 3(1), 57–75. <https://doi.org/10.1007/s40362-015-0031-9>
- Stubbington, R. (2012). The hyporheic zone as an invertebrate refuge: A review of variability in space, time, taxa and behaviour. *Marine and Freshwater Research*, 63(4), 293–311. <https://doi.org/10.1071/mf11196>
- Stubbington, R., England, J., Wood, P. J., & Sefton, C. E. M. (2017). Temporary streams in temperate zones: Recognizing, monitoring and restoring transitional aquatic-terrestrial ecosystems. *Wiley Interdisciplinary Reviews: Water*, 4(4), 4, e1223. <https://doi.org/10.1002/wat2.1223>
- Timoner, X., Borrego, C. M., Acuña, V., & Sabater, S. (2014). The dynamics of biofilm bacterial communities is driven by flow wax and wane in a temporary stream. *Limnology and Oceanography*, 59(6), 2057–2067. <https://doi.org/10.4319/lo.2014.59.6.2057>
- Timoner, X., Buchaca, T., Acuña, V., & Sabater, S. (2014). Photosynthetic pigment changes and adaptations in biofilms in response to flow intermittency. *Aquatic Sciences*, 76(4), 565–578. <https://doi.org/10.1007/s00027-014-0355-6>
- Townsend, C. R., & Hildrew, A. G. (1994). Species traits in relation to a habitat templet for river systems. *Freshwater Biology*, 31(3), 265–275. <https://doi.org/10.1111/j.1365-2427.1994.tb01740.x>
- Vander Vorste, R., Corti, R., Sagouis, A., & Datry, T. (2016). Invertebrate communities in gravel-bed, braided rivers are highly resilient to flow intermittence. *Freshwater Science*, 35(1), 164–177. <https://doi.org/10.1086/683274>
- Vander Vorste, R., Malard, F., & Datry, T. (2016). Is drift the primary process promoting the resilience of river invertebrate communities? A manipulative field experiment in an intermittent alluvial river. *Freshwater Biology*, 61(8), 1276–1292. <https://doi.org/10.1111/fwb.12658>
- Vincent, W. F., & Howard-Williams, C. (1986). Antarctic stream ecosystems: Physiological ecology of a blue-green algal epilithon. *Freshwater Biology*, 16(2), 219–233. <https://doi.org/10.1111/j.1365-2427.1986.tb00966.x>
- Virtanen, R., Muotka, T., & Saksa, M. (2001). Species richness–standing crop relationship in stream bryophyte communities: Patterns across multiple scales. *Journal of Ecology*, 89(1), 14–20. <https://doi.org/10.1046/j.1365-2745.2001.00513.x>

- Vitova, M., Bisova, K., Kawano, S., & Zachleder, V. (2015). Accumulation of energy reserves in algae: From cell cycles to biotechnological applications. *Biotechnology Advances*, 33(6), 1204–1218. <https://doi.org/10.1016/j.biotechadv.2015.04.012>
- von Schiller, D., Bernal, S., Dahm, C. N., & Martí, E. (2017). Nutrient and organic matter dynamics in intermittent rivers and ephemeral streams. In T. Datry, N. Bonada, & A. Boulton (Eds.), *Intermittent rivers and ephemeral streams: Ecology and management* (1st ed., pp. 135–160). Burlington, VT: Elsevier.
- Walters, A. W., & Post, D. M. (2011). How low can you go? Impacts of a low-flow disturbance on aquatic insect communities. *Ecological Applications*, 21(1), 163–174. <https://doi.org/10.1890/09-2323.1>
- Walters, K. R., Sformo, T. J., Barnes, B. M., & Duman, J. G. (2009). Freeze tolerance in an arctic Alaska stonefly. *Journal of Experimental Biology*, 212, 305–312. <https://doi.org/10.1242/jeb.020701>
- Zeglin, L., Dahm, C., Barrett, J., Gooseff, M., Fitzpatrick, S., & Takacs-Vesbach, C. (2011). Bacterial community structure along moisture gradients in the parafluvial sediments of two ephemeral desert streams. *Microbial Ecology*, 61(3), 543–556. <https://doi.org/10.1007/s00248-010-9782-7>

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